

Lowest common ancestor (tag: lca)

Training suite Available memory: 512 MB. Maximum running time: ? s.

You are given a tree. Your task is to answer queries asking for the lowest common ancestor (LCA) of two vertices in the tree.

Input

The first line contains two integers, n ($2 \leq n \leq 500\,000$) and q ($1 \leq q \leq 500\,000$), denoting the number of vertices in the tree and the number of queries.

The second line contains a single integer r , the root of the tree.

Each of the next $n - 1$ lines contains two integers u and v , indicating an edge between vertices u and v .

Each of the next q lines contains two integers x and y , representing a query asking for the lowest common ancestor of vertices x and y .

The vertices are numbered from 1 to n .

Output

Print q lines. For each query, output a single integer: the lowest common ancestor of the two queried vertices.

Subtasks

Subtask	Constraints
1	$n, q \leq 10\,000$
2	$n \leq 10\,000$
3	$q \leq 10\,000$
4	No additional constraints